BA ZC581 :
Applied Data Science

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You can join my unofficial WhatsApp group

A. Follow this link to join my WhatsApp group:

https://chat.whatsapp.com/Lggcdp80BMUIHVKOstn57g

B. Or scan the QR Code to join:

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0.1 Preliminaries

Example Scenario 1:

SimpleLoans Bank offers easy loans to eligible working professionals.

The Applicant is expected to fill an application form and provide details like:

- Name
- Age
- Designation
- Employer Name
- Years of Experience
- Credit Score
- Loan Amount Applied

Can the loan be sanctioned? If so, at what rate?

Bank's Data

Dataset

Borrower ID	Credit Score	Interest Rate (%)
01	500	7.31
02	600	6.70
03	700	5.95
04	700	6.40
05	800	5.40
06	800	5.70
07	750	5.90
08	550	7.00
09	650	6.50
10	825	5.70

Analyze this data

New Data With Unknown Interest Rate

Borrower ID	Credit Score	Interest Rate
11	625	?

To decide this

Example Scenario 2:

- Suppose two wire cut machines produce nails which are on average 10 inches long. A sample of 11 nails is selected from each machine's output.

Machine A: 6, 8, 8, 10, 9, 10, 11, 10, 12, 12, 14

Machine B: 6, 6, 6, 8, 8, 10, 12, 12, 14, 14, 14

- Which machine is performing better ?

How will you do this?

Example Scenario 3:

- Predicting Sales for a Retail Chain
  - Business Goal: Forecast monthly sales for 100 stores to optimize inventory and staff
  - Dataset (snapshot of 2 stores for 2 months):

Store ID	Month	Footfall	Promotions (%)	Sales (₹)
101	Jan	12,000	10	₹18,50,000
101	Feb	11,500	5	₹17,20,000
102	Jan	9,000	15	₹14,80,000
102	Feb	8,200	10	₹13,90,000

  - **How will you do this?**
    - **Apply Data Science Techniques**

Example Scenario 3:

Data Science at Work

- Data Science Techniques Used:
  - Regression Modeling: Predict sales based on footfall and promotion percentage.
  - Feature Engineering: Include holidays, weather, and competitor activity.
  - Model Output Example:
    - Predicted sales for Store 101 in March (footfall: 13,000, promotions: 12%) → ₹19,75,000

Example Scenario 3:

Business Impact:

- **Inventory Optimization:** Reduced Overstock by 18%
- **Staff Scheduling**: Improved shift planning accuracy by 25%
- **Revenue Growth:** Targeted promotions increased monthly sales by ₹1.2 crores across all stores

Do not edit
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What is your core background?

The Slido app must be installed on every computer you're presenting from

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0.2 Course Details

Disclaimer

- The slides presented here are obtained / extracted from the authors of the prescribed books and from various other subject contributors.
- I hereby thank and acknowledge all the contributors for their material and inputs.
- I have added and modified a few slides to suit the requirements of the course.

Course Objectives

No	Course Objective
CO1	Understand the data value chain and its relevance in AI-enabled business decision-making across sectors.
CO2	Develop hands-on proficiency in Python-based tools (Pandas, NumPy, Seaborn, Matplotlib) for handling, cleaning, and analyzing business data.
CO3	Explore internal and external data sources including APIs, databases, and cloud-based repositories, with a focus on structured and unstructured formats.
CO4	Perform comprehensive exploratory data analysis (EDA), feature engineering, and transformation using real-world datasets.
CO5	Build responsible data workflows by applying principles of data governance, compliance, and ethics in business settings.

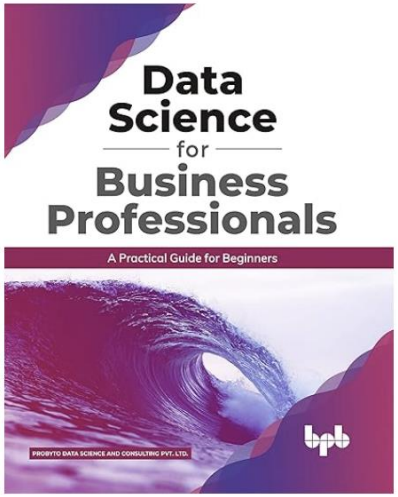
Learning Outcomes

No	Learning Outcome
LO1	Design and execute a complete data preparation pipeline in Python to support AI readiness.
LO2	Conduct EDA to uncover actionable insights and business narratives using Python libraries.
LO3	Apply domain-agnostic data preprocessing techniques, including missing data handling, encoding, normalization, and transformation.
LO4	Integrate structured and semi-structured data from APIs, databases, and cloud platforms.
LO5	Interpret regulatory, ethical, and security aspects of data workflows in compliance with standards such as GDPR and HIPAA.

Text & Reference Books

Text Book:

T1	Probyto Data Science and Consulting Pvt. Ltd. (2020). *Data science for business professionals*. BPB Publications.



Reference Books:

R1	Kakarla, R., Krishnan, S., & Anurag, A. (2020). *Applied data science using PySpark: Learn the end-to-end predictive model-building cycle* (1st ed.). Apress.
R2	Grus, J. (2019). *Data science from scratch: First principles with Python* (2nd ed.). O'Reilly Media.
R3	Provost, F., & Fawcett, T. (2013). *Data science for business: What you need to know about data mining and data-analytic thinking*. O'Reilly Media.

Pre-requisite

You have heard about Data / Data Science / Data Analysis / Data Analytics / Machine Learning / Artificial Intelligence

You understand basic Data operations using MS Excel

You know some programming language and have created a program before, even if small

How will this course be run?

- We will rely on examples and cases in classroom. Exams will also be significantly geared towards case analysis.
- The classes will be interactive (Polls, Quizzes, Q&A, Reading, and Writing) and will include short case studies.
- We will use the LMS portal for continuing the discussions outside the class.
- There will be a mini project to practice what we learn in the class.
- Exams will focus on testing your understanding and thinking. Don't worry if you can't remember specific terms or definitions.
- We will not cover all the book materials in class, so do plan to read the books (or online references) when you need to.
- You need to write an engagement report at the end of the course (your engagement in case responses, Q&A, class polls, questions via form, etc.).

AI usage in this course

1. In general, I encourage you to use AI to learn better. However, use it responsibly, definitely do not cheat using AI!
2. If you use AI in your submissions, declare it – mention the specific purpose you used it for and why.
3. Some submissions may ask you to use AI in a specific part or in a specific way. Please follow such instructions.

Bad use

- Getting direct answers to questions and assignments
- Using AI output as the submission for course work

Good use

- 1-1 tutor to understand concepts discussed in the class
- Generating and brainstorming ideas

Elearn portal details

Elearn portal: https://elearn.bits-pilani.ac.in

- Participants are expected to visit the Elearn portal on a regular basis and stay up to date with all the latest announcements and deadlines.
- Check Taxila/Elearn portal or the MS Teams uploads for all resources and documents. (session recordings, ppts, assignments, case studies, pre reads, add-on material etc)
- WhatsApp Group: An unofficial WhatsApp group would be used to facilitate course relevant communication. Joining Link is announced in the chat window.

Evaluation Scheme

EC	Name	Type	Duration	Weightage	Day, Date, Session, Time
EC-1	Quiz-I	Online	Typically 60 min	5%	TBA
	Quiz-II	Online	Typically 60 min	5%	TBA
	Experiential Learning / Assignment	Activity	Typically 10 days	20%	TBA
EC-2	Mid-Semester Test	Closed Book	2 hours	30%	Sunday, Dec 21st, 2025 (R) Sunday, Jan 4th, 2026 (M) Time: TBA
EC-3	Comprehensive Exam	Open Book	2.5 hours	40%	Sunday, March 1st, 2026 (R) Sunday, March 8th , 2026 (M) Time: TBA

*** EC = Evaluation Component** **** R = Regular Exam** ***** M = Makeup Exam**

Syllabus for Mid-Semester Test: Topics in Contact Session Nos. 1 to 8 **

Syllabus for Comprehensive Exam: All topics in Session Nos. 1 to 16 (all topics)

**** typically covers up to module 8 from handout. Any change will be notified.**

Experiential Learning / Situated Learning Project

As part of the EC1 evaluation component, you are expected to complete a situated learning project.

Coding will be using Low Code / No Code tools

Details will be shared in the class at the appropriate time

Grading

- Course weightage: **4 credits (3-0-1)**
- Grading Model: Relative Grading
- Grading:
  - Total marks obtained (out of 100) is used to prepare the **distribution bell curve**. Using the class average and the corresponding spread (standard distribution), grades (A, A-, B, B-, C, C-, D, and E) are assigned.
  - Typically, A is awarded to the top 2.5-3.5 percentile students of the class.
  - An E grade would be a RLE / RRA (repeat the course)

No Unfair practices please. It may lead to a disqualification of the subject/semester

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Let's Get Started. Welcome to ADS!

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Module 1: Introduction to Applied Data Science

Module 1 Topics

Topic 0: Introduction to Data Science

Topic 1: Understanding the Data Value Chain

Topic 2: AI Readiness

Topic 3: Role of Data Science in Business

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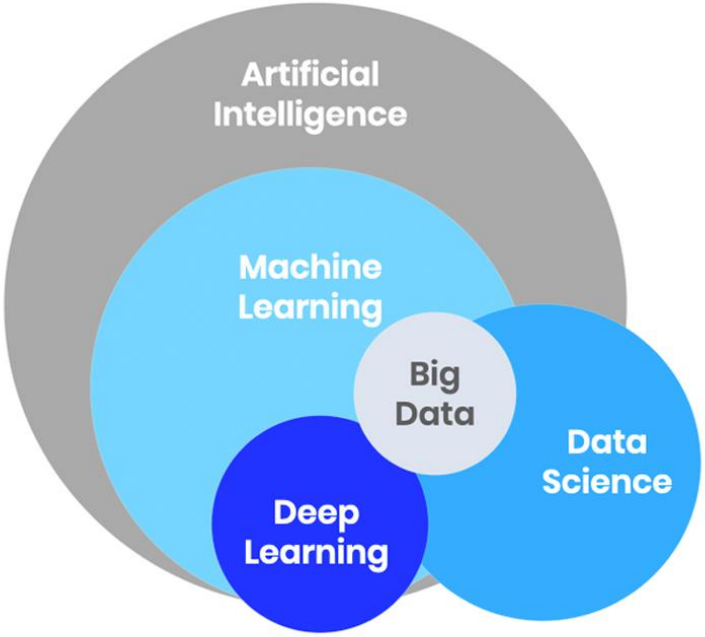
Topic 0: Introduction to Data Science

What is Data Science

Definition, evolution and relevance in AI and business



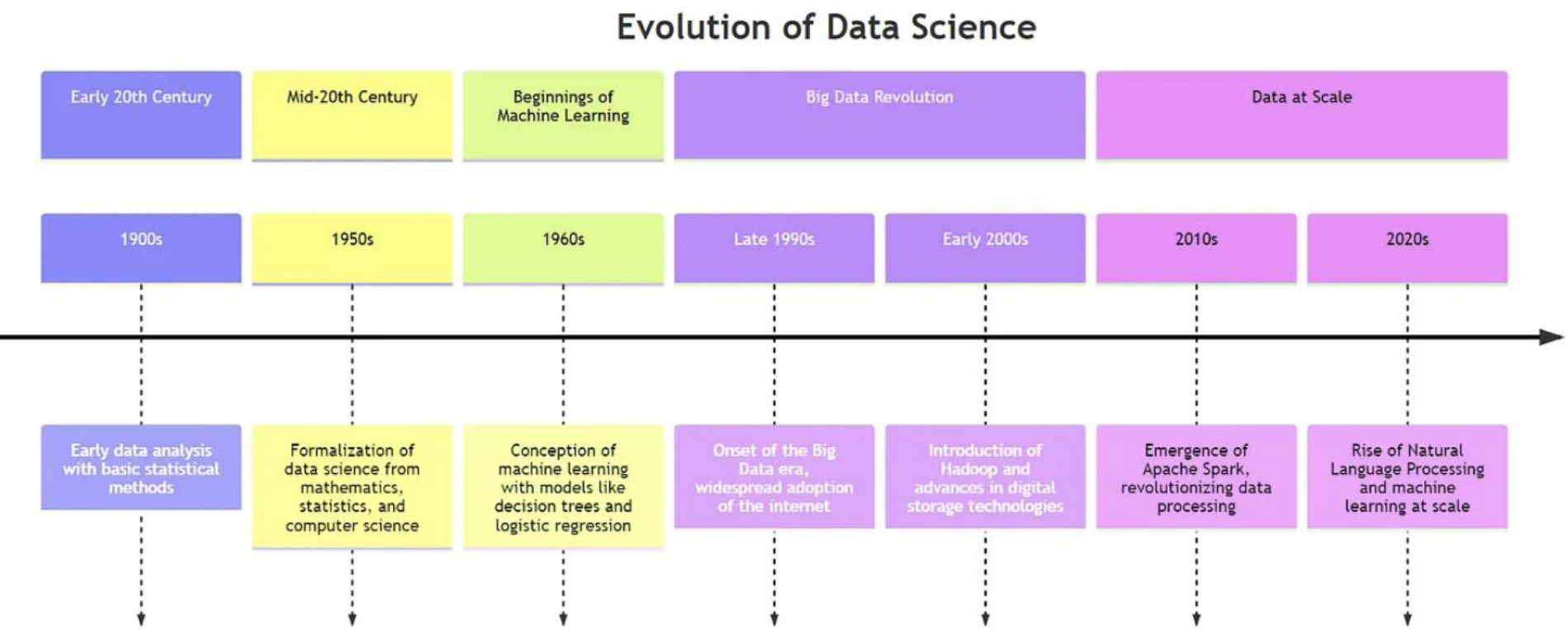
What is Data Science?

- Data Science is a compilation of techniques that extract value from data.
  - Value of stored business data is zero unless it is acted upon.
- An essential tool for any organization that collects, stores and processes data as part of its operations.
- Techniques rely on finding patterns, connections and relationships within data.
- Some techniques used include
  - Applied Statistics
  - Machine Learning
  - Visualization
  - Logic

Data Science – A definition

- Data **Science** is the study and application of methods to analyze large volumes of data—structured or unstructured—to uncover patterns, make predictions, and support decision-making.
  - It blends mathematics, statistics, programming, and domain knowledge to solve real-world problems across industries like healthcare, finance, retail, and technology.
- Other terms that sound similar: Data Analytics, Data Analysis

Evolution of Data Science



Growth in Data Science: Facts & Figures

Amount of data generated is 175 Zettabytes by 2025

Data Science Platform market will be worth 1830B by 2033 from 145B today

Growing at a CAGR of 28.8%

Demand for Data Scientists increased by 56% from 2020-2022

Average salary in US is $125K

65% of organizations believe that Data science is essential for decision making

90% believe DS is crucial for business success

Python is most popular programming language (66% usage)

ML and DL skills are top sought skills

36% of organizations have implemented AI in businesses

Relevance in Business & AI

Data as a fuel for AI: Data Science transforms raw data into structured insights that power Machine Learning and AI models

Smarter Decision Making: Businesses use DS to forecast trends, optimize operations and personalize customer experience

Predictive Power: Predictive analytics help businesses anticipate the outcomes and act proactively

Efficiency and Automation: AI systems trained on data science pipelines automate repetitive tasks, reduce errors and scale operations

Competitive Advantage: Enables companies to outperform peers in agility, innovation and customer retention

Real-time responsiveness: Businesses can respond instantaneously to market shifts and customer behavior

Why Data Science Matters?

Offers Strategic value

Provides Decision making capability

Fuels Innovation

Data Science vs Traditional Analytics

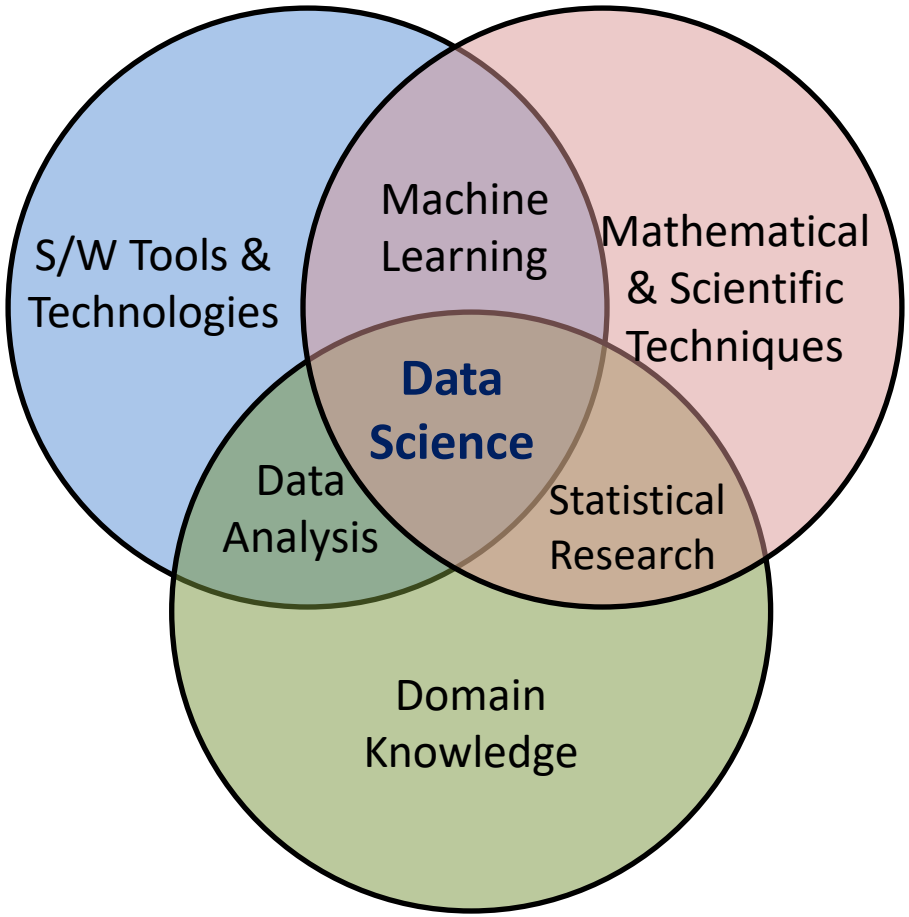
Key Differences:

Aspect	Traditional Analysis	Data Science
Approach	Rule-based, manual interpretation	Algorithmic, automated, predictive
Tools Used	Excel, SQL, basic stats	Python, R, ML libraries, cloud platforms
Data Volume	Small to medium datasets	Big data, unstructured and streaming data
Output	Reports, summaries	Models, forecasts, real-time decisions
Skills Required	Business acumen, basic analytics	Programming, statistics, machine learning

Overlap:
Both rely on Data Cleaning, Exploratory Analysis and Domain Knowledge to support business decision support.

Modern Expectations:
Shift to predictive & Prescriptive Analytics, Integration with AI systems, Support Automation, and offer scalable real time insights.

Foundation Knowledge Required



Example Expertise Needed

- **Mathematical & Scientific Techniques**
  - Linear Algebra
  - Classical Statistical techniques
  - Clustering and Classification
  - ML
- **Tools & Programming**
  - Programming Languages (C++, Java, Python)
  - Analysis Tools (R, SAS, Matlab)
  - Visualization Tools (Tableau, PowerBI)
- **Domain Knowledge**
  - Engineering
  - Science
  - Business
  - Economics
  - Finance
  - Medicine

Scenario:

A bank has to Offer Loan to a Loan Applicant.

- Domain Knowledge (provides business context):
  - What aspects to consider to issue a loan?
  - What is the risk associated with the recovery?
  - What should be the terms and conditions?
  - What is the present market condition?
- Mathematical & Scientific techniques(provides a behavioral model of the business):
  - Risk associated with recovery is HIGH is inferred from domain knowledge. Risk calculated by mathematical techniques says it is 69%
- Tools & Technology(offers a means to evaluate business data for decision making purpose):
  - Use Python, R, Knime to evaluate a model and derive the outcome

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Are you using Data Sciences / Machine Learning / AI at your workplace?

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Topic 1: Understanding the Data Value Chain

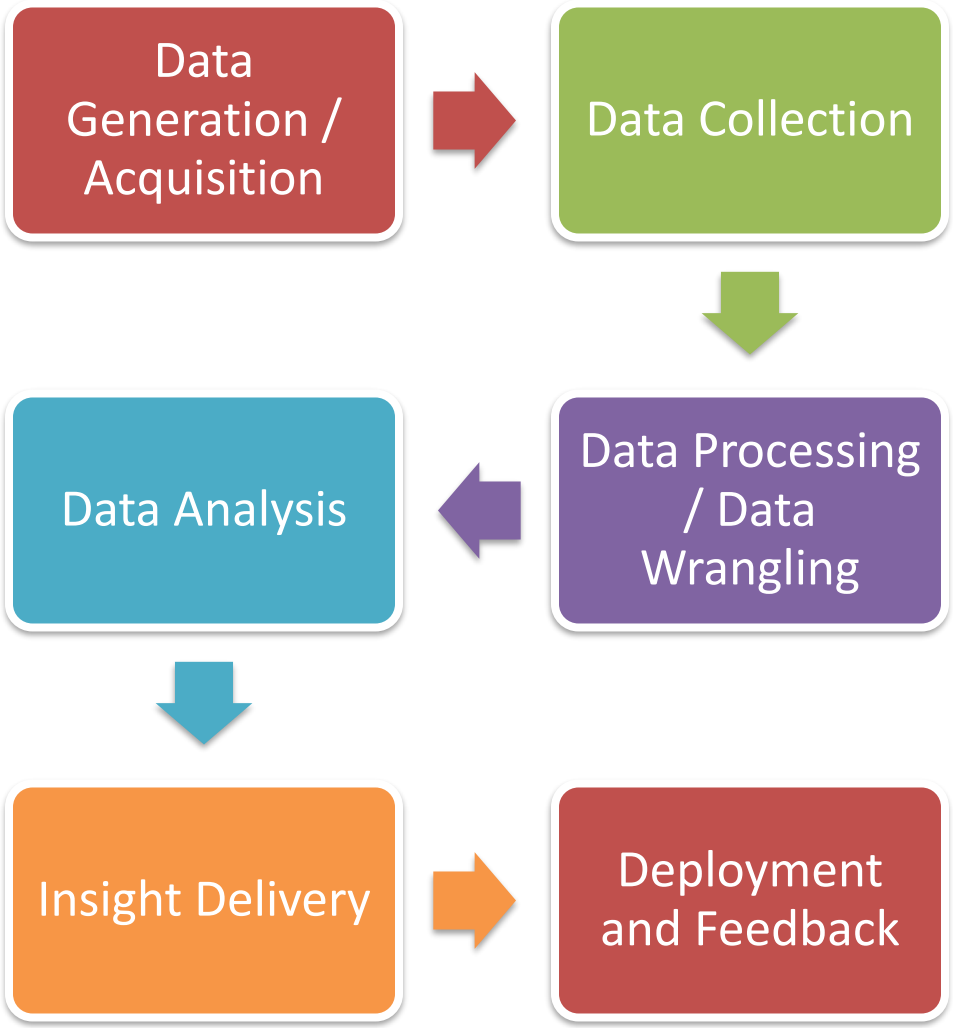
Overview of the Data Value Chain

- **The Data Value Chain:** It maps the journey from raw data to actionable insights, enabling AI and business decisions.

Stage	Description
Data Generation / Acquisition	Sensors in real time process, business transactions, social media postings, system and application event logs, user interactions
Data Collection	Gathering data into databases, data lakes or cloud platforms
Data Processing	Cleaning, transforming and organizing data for analysis
Data Analysis	Applying statistical methods, ML models and visualization tools
Insight Delivery	Produce Reports, dashboards, predictions and decision support
Deployment and Feedback	Operationalizing models and refining based on real-world outcomes

- **Business Impact:** Drives Smarter Decisions, Powers AI systems, Enables Competitive Advantage.

The Data Analysis Pipeline



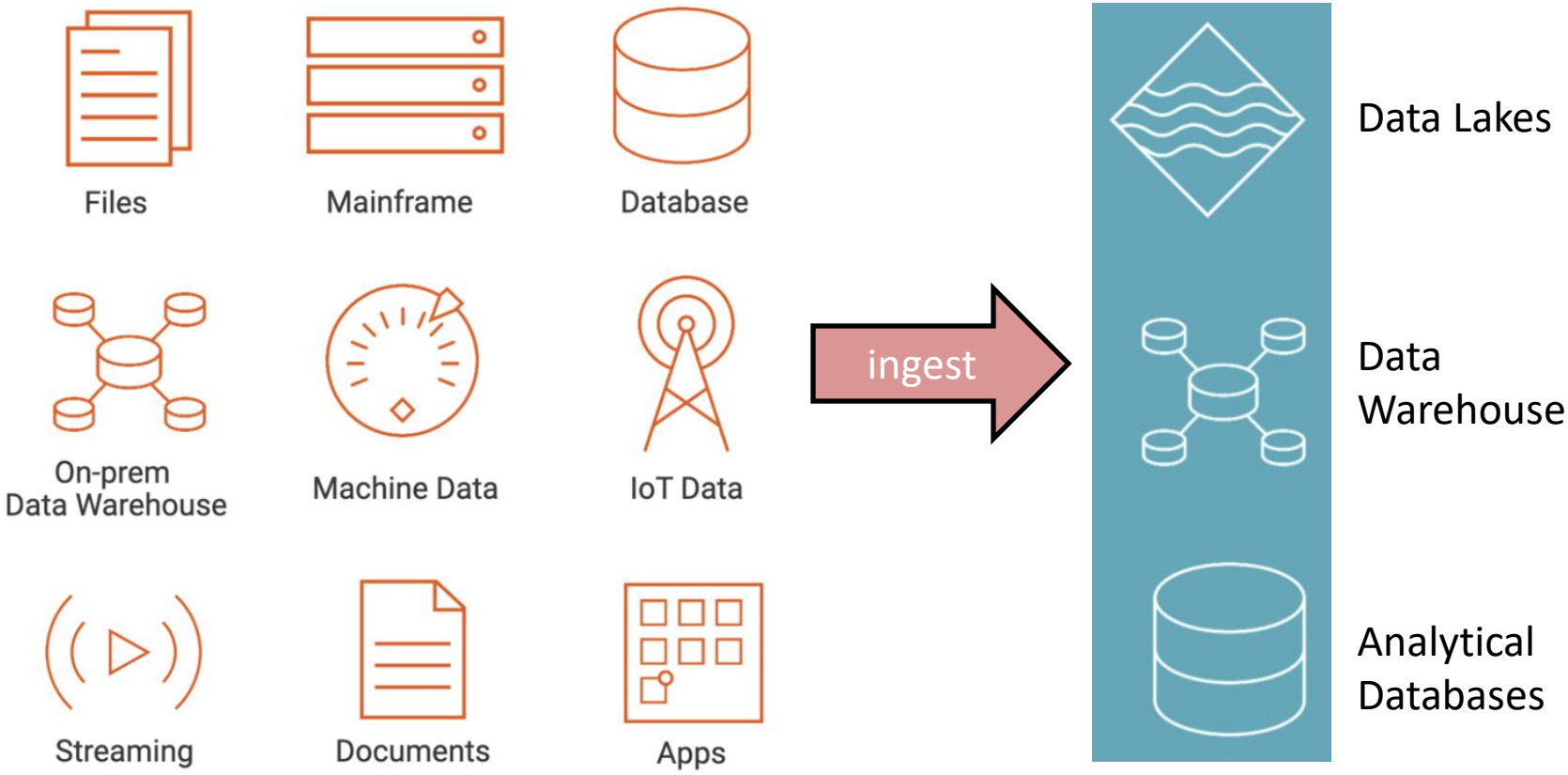
Data Generation

Data ingestion from various sources.

Sources: Sensors, transactions, social media, logs

- Transactional Systems:
  - Sales records, Purchase records, billing logs, POS systems
- Sensors and IoT Devices:
  - Temperature, motion, GPS, industrial and process control equipment
- User Interactions:
  - Clicks, searches, app usage, social media posts
- Business Operations:
  - Inventory updates, employee check-ins, customer feedback
- External Feeds:
  - Weather APIs, Financial market tickers, public datasets
- Synthetic Data:
  - Synthetic / Dummy data generated that resembles real world data with private and confidential data masked with dummy values (data obfuscation).

Data Ingestion



Data Collection

Use methods like

- **Batch uploads:** Periodic data dumps from various systems
- **Streaming Pipelines:** Real time data ingestion from business processes.
- **Web scraping:** Automatic extraction of data from web sites / web resources
- **Manual entry:** Manually enter data using structured forms or spreadsheets

Data Processing & Cleaning

This stage transforms raw, messy data into structured, usable formats for analysis and modeling.

Enhances accuracy, consistency and reliability.

Also called ETL, Data Wrangling

Used for Quality checks

Data Processing & Cleaning

Activity	Meaning
Handle Missing Values	Fill, drop, or impute missing entries to avoid bias
Removing Duplicates	Eliminate repeated records to maintain data integrity
Standardizing Formats	Uniform date formats, currency units, categorical label
Outlier Detection	Identify and treat extreme values that may skew results
Data Transformation	Normalize, scale, encode, or aggregate data for modeling
Noise Reduction	Filter irrelevant or erroneous entries (e.g., typos, sensor glitches

Data Analysis & Modeling

Use a combination of

- **Statistical Methods**
  - Linear Algebra
  - Distribution
  - Correlation
  - Probability
  - Multivariate calculus
  - Optimization
- **Machine Learning Techniques**
  - Regression
  - Classification
  - Clustering
  - Market basket analysis (Association Analysis)
- **AI**
  - CNN & Deep Learning Models
  - Reinforcement Learning Models
  - NLP
  - Adversarial Networks
  - Language Models

Insight Delivery

A process of **translating data analysis into actionable outputs** using techniques such as

Dashboards and Reports

- Visual summaries for executives, analysts, stakeholders and teams

Predictive Models and Alerts

- Forecasts, risk scores, and real-time notifications

Decision Support Systems

- AI-driven recommendations embedded in workflows

Business Presentations

- Storytelling with data for strategic alignment

APIs and Embedded Outputs

- Feeding insights into apps, websites, or automated systems

Deployment & Feedback Loop

Involves **operationalizing data models** into real-world systems and continuously improving them based on performance and user feedback

- **Deployment**:
  - Release for consumption
    - On prem infrastructure or Cloud Infrastructure
  - Model Integration
    - Embed predictive models into apps, dashboards, or decision systems
  - Automation
    - Trigger actions (e.g., alerts, recommendations) based on model outputs
  - Monitoring
    - Track accuracy, latency, and business impact in production
- **Feedback Loop:**
  - User Feedback
    - Ratings, Functional Corrections, Domain Corrections, Behavioral indicators
  - Performance Metrics
    - Error rates, ROI tracking
  - Retraining & Updates
    - Refresh models with new data to keep relevance

Do not edit
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What do you think is the biggest challenge in adopting data science in business?

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Roles in Data Science

Roles in Data Science

1. Data Scientist
2. Data Engineer
3. ML Engineer
4. Business Analyst
5. Cross Functional Collaboration Teams

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Types of Analytics

Types of Analytics

Descriptive

Predictive

Diagnostic

Prescriptive

Inferential

Exploratory

Cognitive

We will study these in detail

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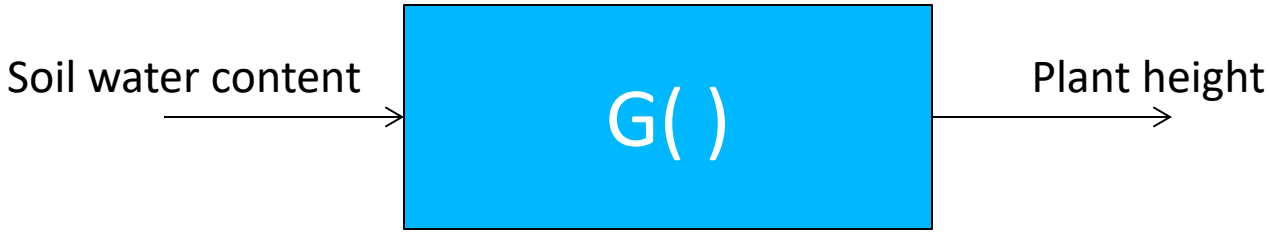
Model Development Process

What is a model?

- Defines the behaviour of a phenomenon / process / activity
- Identifies a mathematical representation of the phenomenon
- Enables us to understand the process
- Establishes the cause and effect relationship

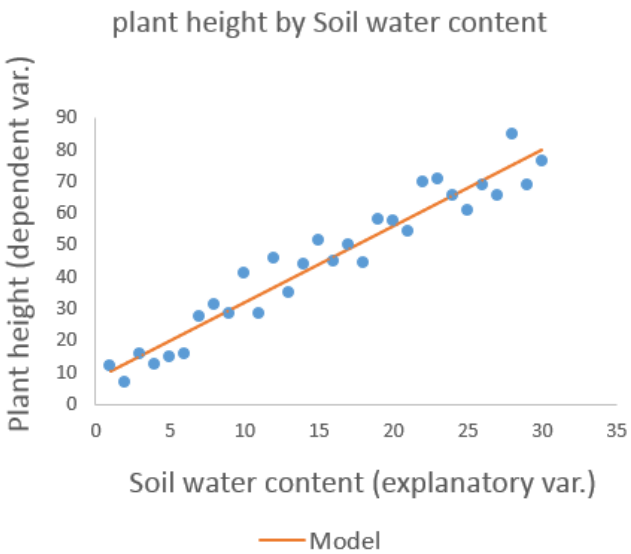
Model
Ex: Growth model of a plant

Representation of a certain phenomenon



Capture observations and algorithm prepares the model

h = plant height
w = water content



Linear model
h=b1*w + b0

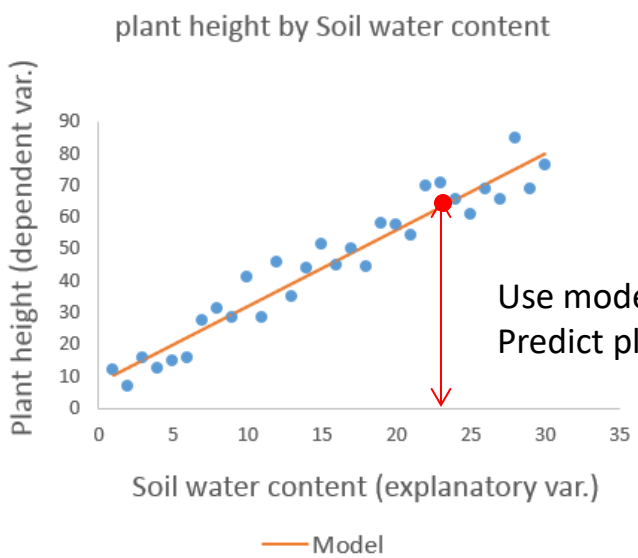
Model
Ex: Growth model of a plant

Representation of a certain phenomenon

Water content → **h**=G(w)=b1*w+b0 → Plant height

Use the model to predict

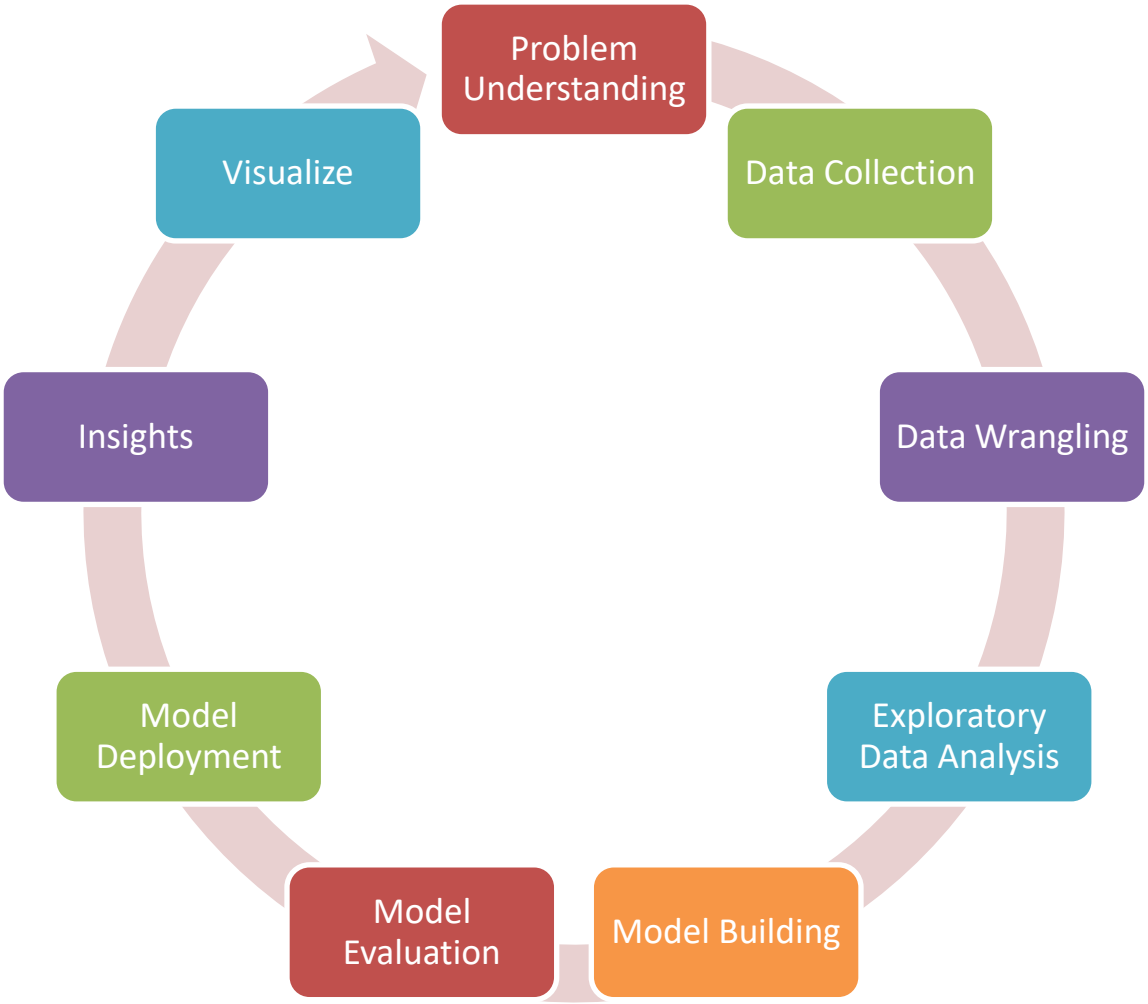
h = plant height
w = water content



Linear model
h=b1*w + b0

Use model
Predict plant height for water content = 23

ML Model Development Process



ML Model Building Process

- Problem Definition: Define objective of the analysis
- Data Collection: Gather needed to solve the problem is data from various sources
- Data Wrangling: Cleaning, transforming and enhancing data quality to make it ready for analysis
- Exploratory Data Analysis: A pre model building analysis to understand distribution and data relationships to assess the best algorithm to use.
- Model building using Machine Learning Algorithms: Use data and train / test and validate a model from a set of models (supervised, unsupervised and reinforced learning models)

ML Model Building Process

- Model Evaluation: Estimate the accuracy and errors of the model's functioning.
- Model Deployment: Integrate the model in an application and release the model to users. Provide suitable infrastructure for model's run time.
- Insights: Derive measurable values for a query and produce outcomes that are actionable.
- Visualize: Present the data in a dashboard or a suitable form for easy consumption.
- Maintenance: Use new data and refine / rebuild the model. Ensure model behavior is as per user need

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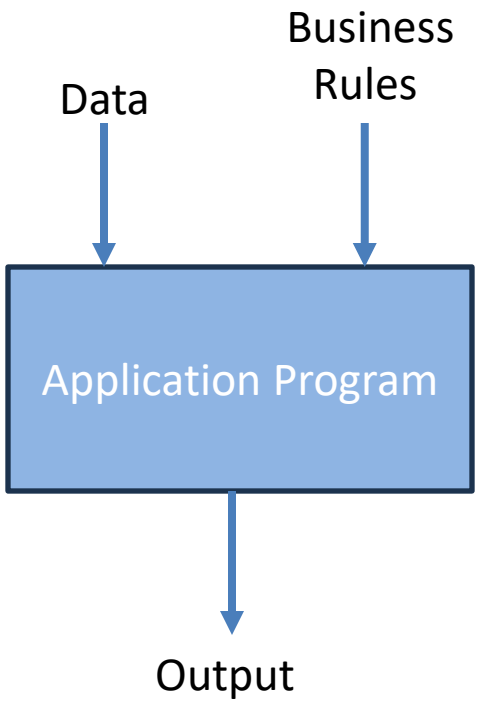
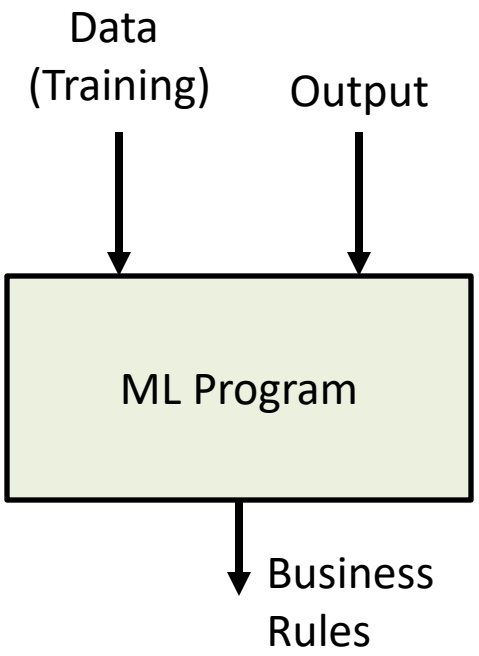
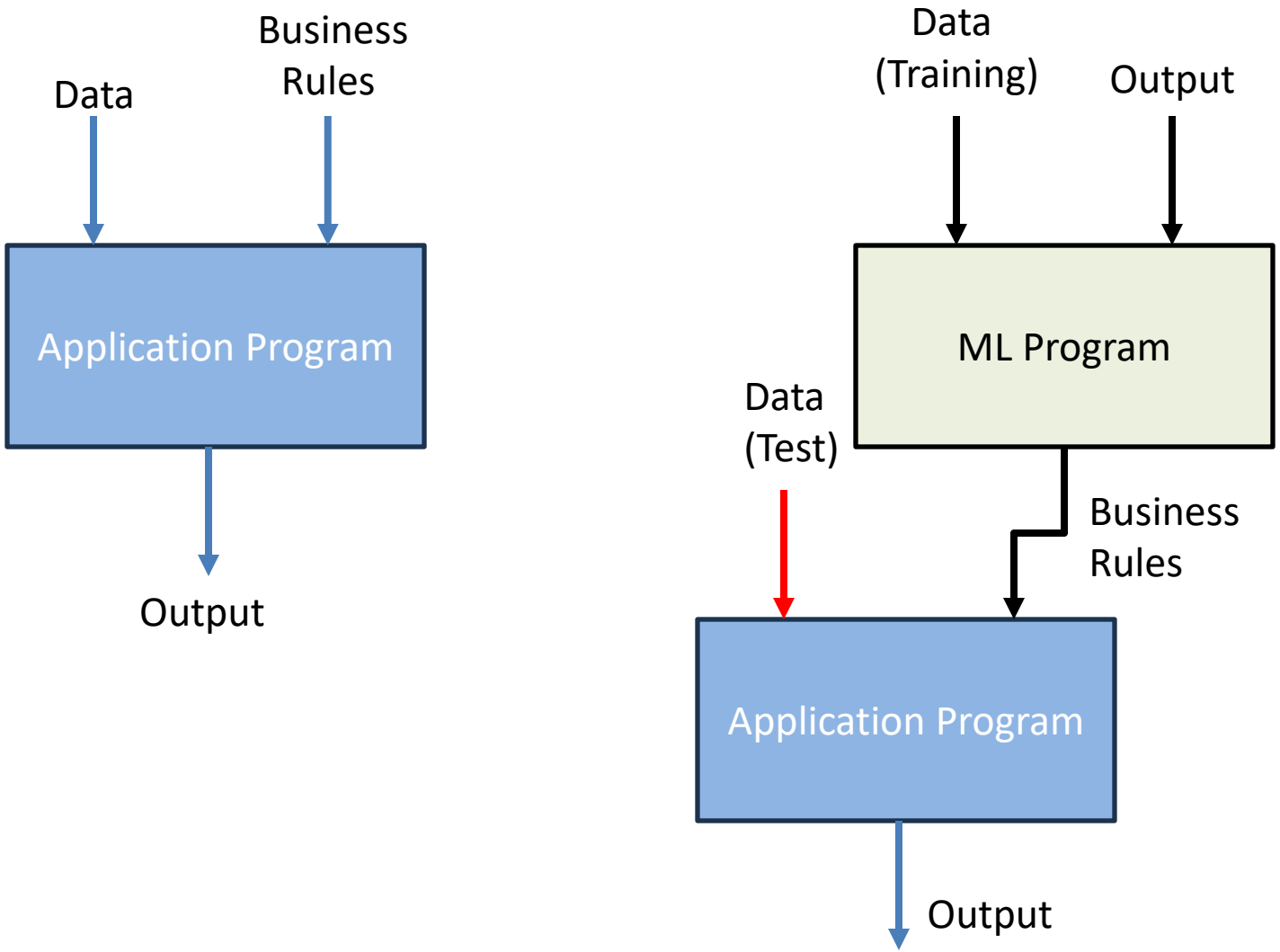
Machine Learning Models

Standard Application vs ML Application





Standard Application vs ML Application

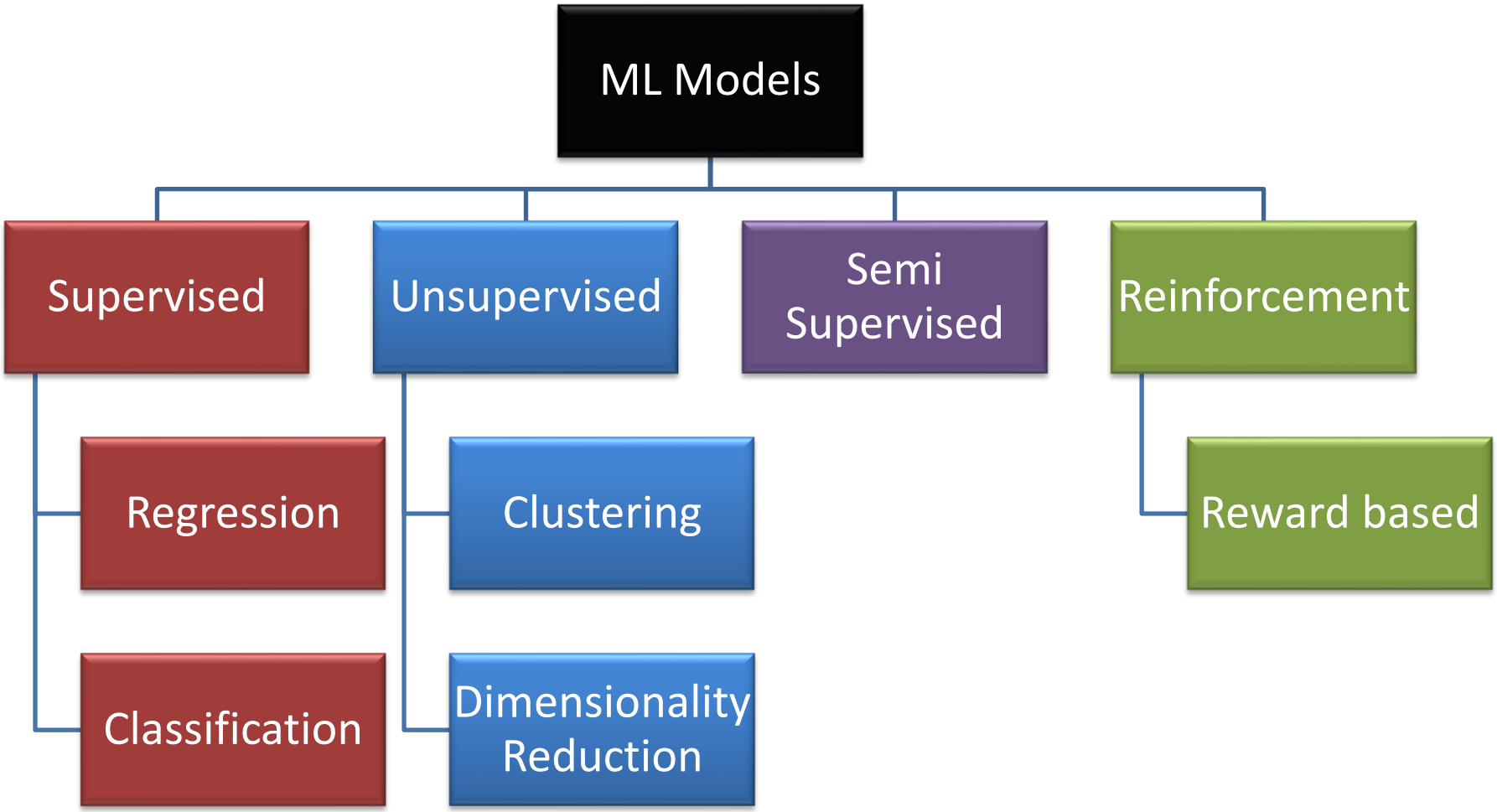


ML Models

Machine Learning = An algorithm that optimizes performance by learning from examples without explicitly being programmed. It learns from data.

Machine Learning models generate the rule base (program) by learning from data, which can be used in a traditional system to generate outputs on new unseen data.

Learning Methods



Types of ML Models

Supervised Learning Models: The model learns from labeled data (input-output pairs data that is already seen). Goal is to Predict outcomes for new, unseen data.

Unsupervised Learning Models: The model finds patterns or groupings in data without labels. Uses unseen data. Goal is to detect hidden structures or clusters.

Semi Supervised Learning Models: Combines a small amount of labeled data with a large amount of unlabeled data. Improves learning accuracy when seen labeled data is scarce.

Reinforced Learning Models: Learns by interacting with the environment to earn rewards. Goal is to maximize rewards.

Examples of ML Algorithms

Algorithm Category	Examples
Supervised Learning	Linear / Non-linear Regression Decision Trees Naïve Bayes Support Vector Machines KNN Neural Netwoks
Un Supervised Learning	K-means clustering DB Scan Association Rules Principle Component Analysis
Semi Supervised Learning	Self Training Graph based
Reinforced Learning	Temporal Differences Q Learning Adversarial Networks

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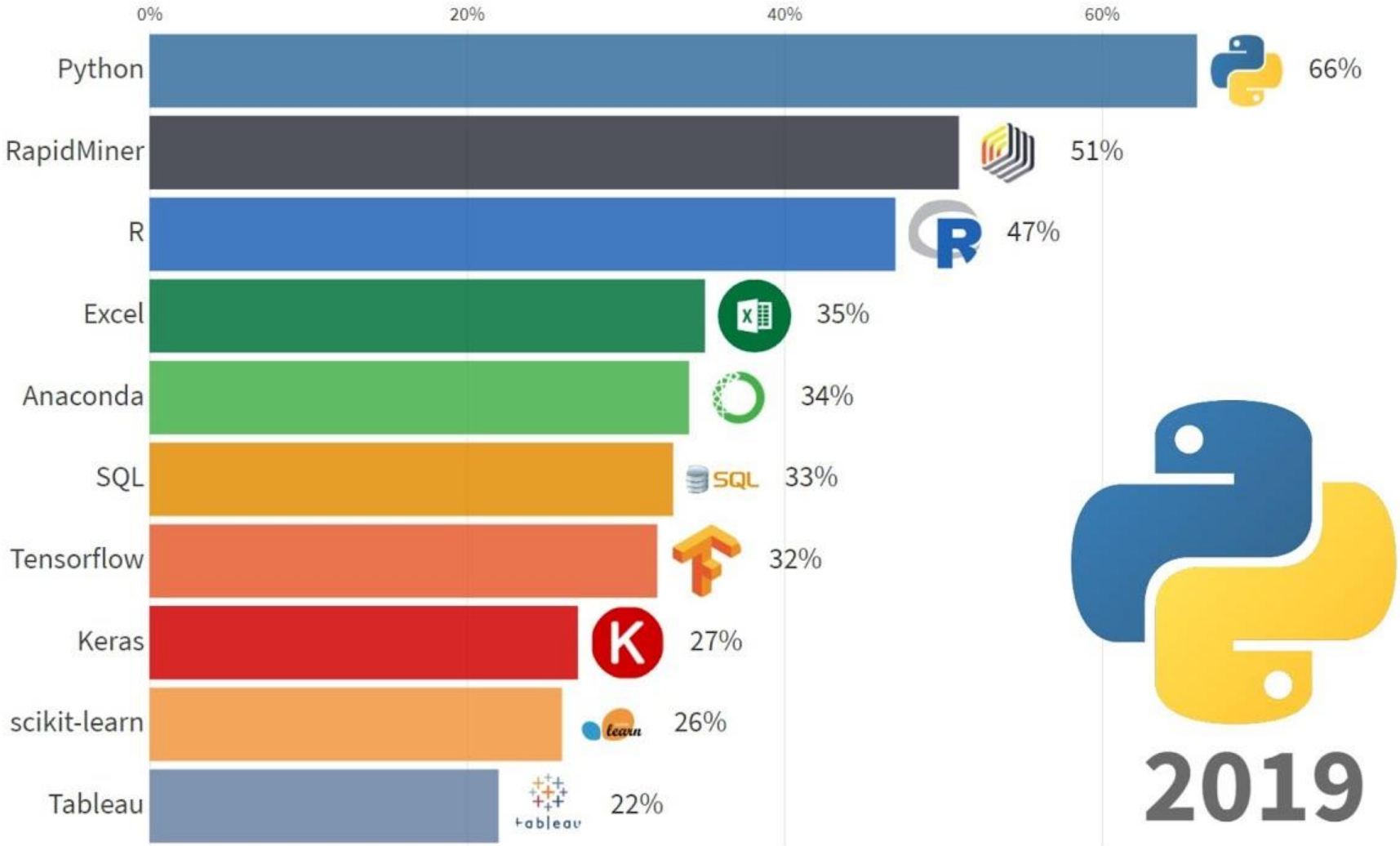
Tools & Technology

Tools & Technology

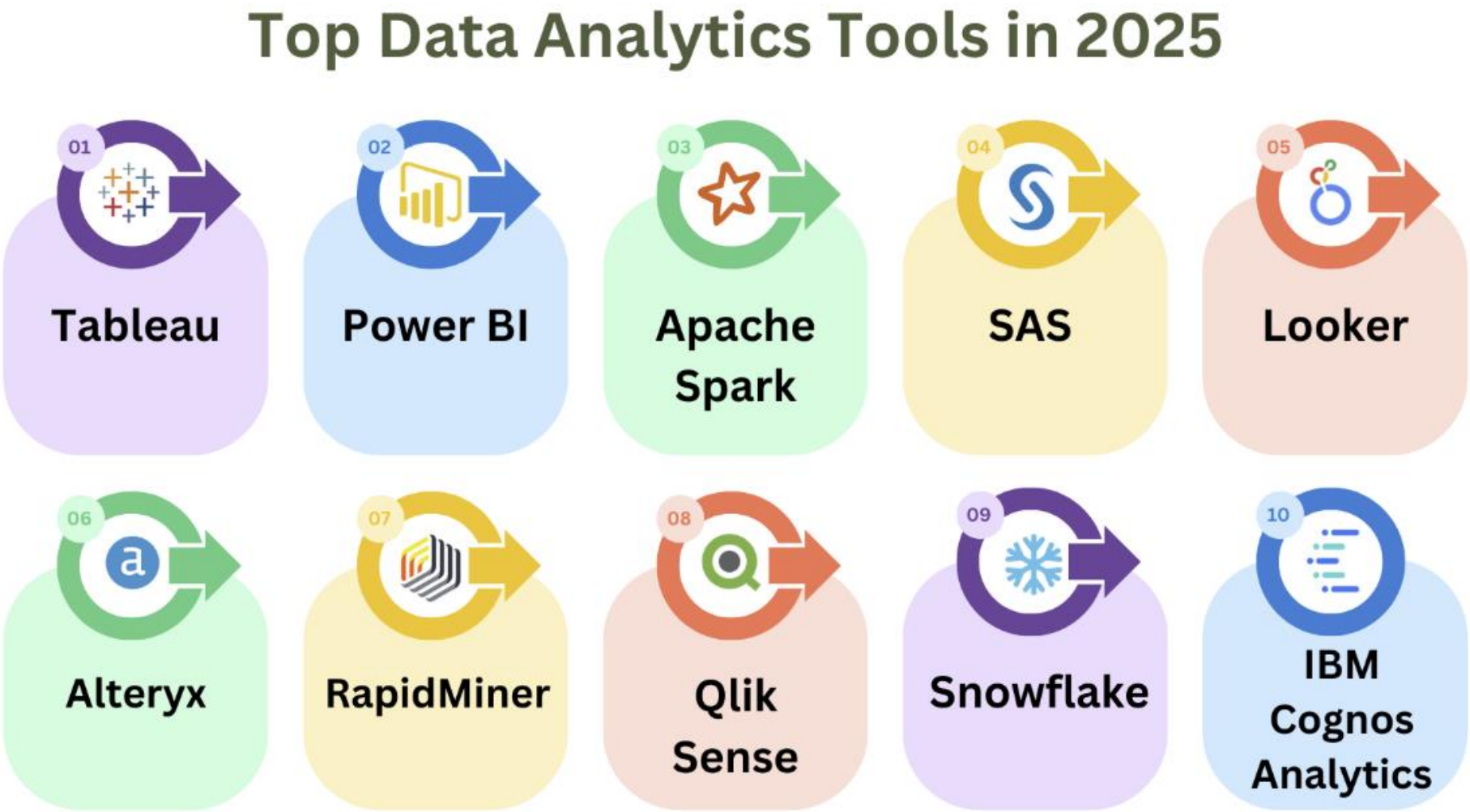
Need appropriate tools and technology to implement solutions at scale

Data Science Professionals should be aware of aware of latest tools and platforms to ensure they realise the most efficient and economic solution.

KDnuggets Analytics, Data Science, ML Software Poll, 2019



Data Analytics tools for 2025



Top Data Science Trends of 2024

1	Generative AI Expansion and Foundation Models	2	Multi-modal Data-driven Solutions
3	AI-fueled Data Automation	4	Explainable AI (XAI) and Accessible Data Tools
5	Ethical and Responsible AI/ML	6	Augmented Analytics and Citizen Data Science
7	Sustainability-focused Data Science Innovation	8	Data Product Management
9	Edge and On-Device AI/ML	10	Quantum Machine Learning

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Topic 2: AI Readiness

What is AI readiness?

Definition: The ability of an organization or system to adopt, integrate and scale AI solutions in their operations

Key Dimensions:

- Strategy
- Infrastructure
- Data
- Talent
- Culture

Why AI Readiness Matters?

It offers

- Competitive Advantage
- Operational Efficiency
- Innovation and customer experience
- Risk mitigation

Strategic Alignment

Clear AI Vision and Goals

Leadership buy-in

Integration with business strategy

Building a Data Driven Culture

Leadership

Literacy

Ethics

Data Infrastructure

Storage and Processing Infrastructure

- On prem
- Hybrid
- Off prem (Cloud)

Data Availability, quality and governance

Storage, pipelines and access

Privacy and compliance readiness

Technology Stack

Cloud Platforms , APIs and compute resources

Dev teams, dev tools

ML Ops and deployment tools

Cybersecurity and scalability

Talent & Skill

In-house capabilities: data science, ML engineering, domain expertise

Upskilling and training programs

Collaboration with external partners

Culture and Change Management

Openness to experimentation

Cross-functional collaboration

Ethical awareness and responsible AI mindset

Assessment Framework

AI readiness scorecards or maturity models

Gap analysis and roadmap creation

KPIs and success metrics

Next Steps

Pilot Projects

Governance Setup

Continuous Learning and Iteration

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Topic 3: Role of Data Science in Business

Intersection of Data Science, AI and Business

Industry adoption is happening

- How Data science powers AI
- Data as fuel for Machine Learning
- Training, validation and inference cycles
- How AI transforms business decision-making

How Data Science Powers AI

1. Data Science as the Engine of AI

- Data science provides the tools, techniques, and workflows to extract insights from data.
- It enables AI systems to learn, adapt, and make decisions using statistical models and machine learning.

2. Data as Fuel for Machine Learning

- Raw data (images, text, transactions, sensor readings) is the input that powers ML algorithms.
- The quality, volume, and diversity of data directly impact model accuracy and reliability.

AI in Business

1. How AI Transforms Business Decision-Making

- AI enables real-time, data-driven decisions across marketing, finance, HR, and operations.
- Examples:
  - Personalized product recommendations in e-commerce
  - Fraud detection in banking and credit card useage
  - Predictive maintenance in manufacturing
  - Dynamic pricing in retail

Some Example Application Areas

Sales: Demand Forecasting; Discount offerings; Product bundles, Customer segmentation

Production: Prediction of Downtime; Productivity Analysis; Anomaly detection; Monitoring Systems; Inventory Management & demand forecasting

Quality Control: Inspection Analytics based fault identification; (weld quality)

Logistics & Transportation: Route

Procurement: Vendor rating analysis

Conversational: ChatBots;

THINK OF USING WHEN YOU HAVE LARGE DATA TO WORK WITH

Broad Domains

RETAIL

SALES, MARKETING & MEDIA

PROCUREMENT

FINANCE

MANUFACTURING

HEALTHCARE

TRAINING & EDUCATION

GOVERNMENT & POLICY

Future Trends

Generative AI

Synthetic Data

Responsible AI

Responsible and Ethical AI

- Responsible AI refers to the ethical and transparent development and deployment of artificial intelligence technologies.
- Designed to mitigate bias, ensure privacy, and prioritize the well-being of all users.
- Responsible AI ensures that algorithms and models are designed to make ethical decisions, reducing harm. It protects personal data, addressing privacy concerns and complying with data protection laws.
- Responsible AI complies with existing laws, like GDPR in Europe or CCPA in California. Non-compliance may result in hefty fines and damage to a company's reputation

Summary

- Discussed the importance of IS in an organization
- Identified the Business roles of an IS
- Studied Information System Model and IS Components
- Identified IS Resources and IS Activities

Questions?